POSSESION OF MOBILES IN EXAM IS UFM PRACTICE.

Name _____ _____ **Enrollment No.**_____ _____

Jaypee Institute of Information Technology, Noida
T2 Examination, 2024
B.Tech. VI Semester

Course Title: Computer Networks and Internet of Things **Maximum Time: 1 hr.**
Course Code: 18B11CS311 **Maximum Marks: 20**

CO1	Defining the basics of networking, components and underlying technologies.
CO2	Illustrate the various key protocol in OSI model and TCP/IP protocol suite and explain various application protocols.
CO3	Identification and description of various components, architectures and protocols of internet of things (IoT) and their real-life problems.
CO4	Choose the appropriate network layer and datalink layer protocols based on the specific requirements of the communication environment.
CO5	Explain various transport protocols and its performance enhancing mechanisms.
CO6	Determine the shortest path for the network using various routing protocols and evaluate it.

1. Based on the following parameters, compare and contrast Selective Repeat with Go-Back-N sliding window protocols: **[CO5 (Analysing), 3 Marks]**

   i. How is the semantic of the Acknowledgements being different?
   ii. How is the semantic of timeout being different?
   iii. How is the amount of timers required at the sending node being different?
   iv. How is the memory requirement at the receiving node being different?
   v. How is the amount of packet retransmission being different?
   vi. Suppose we know that most of the packet losses are due to congestion on the internet and that when there is congestion, consecutive packets tend to be dropped altogether due to buffer overflow at the intermediate routers. Which of Selective Repeat and Go-Back-N is better for the internet? And why?

2. **[CO5 (Analysing), 2 Marks]** Suppose that TCP's current smoothed values for the round trip time (RTT_S) and deviation in the RTT(RTT_D) are 400msec and 12msec, respectively. Suppose that the next measured value of the RTT is 280msec. Compute TCP's new value of RTT_S, RTT_D, and TCP timeout value. Use the values of $\alpha = 0.125$, and $\beta=0.25$.

   [CO5 (Analysing), 2 Marks]

3. Consider a network segment containing an IPv4 packet with a total length of 2000 bytes and identification number 56432. It is required to fragment this packet to adhere to a maximum transmission unit (MTU) along the path.

   i. Given an MTU of 1500 bytes, calculate the HLEN, total length, identification number, flags, fragmentation offset fields for all the fragments.

   [CO4 (Applying), 3 Marks]

ii. Now, assume the second fragment obtained from the previous step encounters another network with a reduced MTU of 500 bytes, necessitating further fragmentation. Calculate the HLEN, total length, identification number, flags, fragmentation offset fields for the new fragments resulting from this second-level fragmentation.

[CO4 (Applying), 3 Marks]

4. Suppose TCP Reno sender 'Alice' wants to send total 14 segments to 'Bob'. Rwnd and Cwnd initially at the start of 1 MSS and threshold at the start of first transmission is 8 MSS. From 2nd round onwards the receiver window size start is 1,2,4,2,2,5,3,1,4. A time-out occurs for segment 11 and three duplicate ACKs are received for segment 14. Show the entire evaluation process of congestion window as each segment is sent.

   i. In which transmission round the retransmitted 11th segment is being sent.

   [CO5 (Analysing), 2 Marks]

   ii. What would be the time when the 14th segment is being retransmitted (RTT=10msec)?

   [CO5 (Analysing), 2 Marks]

   iii. What is the Rwnd and Cwnd size when 14th segment is being transmitted.

   [CO5 (Analysing), 2 Marks]

5. Let us consider a message bit Message=110101 that is to be transmitted using CRC, the generator polynomial is x^2+1. **[CO4 (Applying), 3 Marks]**

   i. Calculate the CRC value and the data bit that is sent by the sender to the receiver.

   ii. At, the receiver end, how it will calculate, whether there is any error in the data bits received or not. Is there any error reported by receiver or not?